

WUMPUS LOGIC AGENT

PROPOSITIONAL INFERENCE ENGINE — KNOWLEDGE-BASED REASONING

ENVIRONMENT CONFIGURATION

ROWS

4

COLS

4

PITS

3

GENERATE WORLD



AGENT



SAFE



PIT / WUMPUS



UNKNOWN



VISITED



WUMPUS WORLD MAP

4,1 ?	4,2 ?	4,3 ?	4,4 ?
3,1 ?	3,2 ?	3,3 ?	3,4 ?
2,1 ?	2,2 ?	2,3 ?	2,4 ?
1,1  AGENT	1,2 ?	1,3 ?	1,4 ?

AGENT AT [1,1] — EXPLORING THE DUNGEON...

Click an adjacent cell to move the agent. Agent starts at [1,1] (bottom-left).

AUTO STEP

REVEAL ALL



REAL-TIME METRICS

INFERENCE STEPS

4

CELLS VISITED

1

SAFE DEDUCED

2

DANGER DEDUCED

0

KB CLAUSES

7



CURRENT PERCEPTS

No percepts at [1,1].

KNOWLEDGE BASE LOG

FACT: $\neg P_{1,1}$ (start is safe)

VISIT [1,1]:

TELL: $\neg B_{1,1}$ (No breeze)

TELL: $\neg S_{1,1}$ (No stench)

INFER: $\neg P_{2,1}$ (no breeze $\rightarrow$ safe)

INFER: $\neg P_{1,2}$ (no breeze $\rightarrow$ safe)

INFER: $\neg W_{2,1}$ (no stench $\rightarrow$ no wumpus)

INFER: $\neg W_{1,2}$ (no stench $\rightarrow$ no wumpus)

INFERENCE ENGINE LOG

⊥ Contradiction found in 1 steps!

✓ SAFE DEDUCED: [2,1] — $\neg P \wedge \neg W$ proved via resolution

Resolution: proving $\neg P_{3,1}$... (step 3)

⊥ Contradiction found in 1 steps!

Resolution: proving $\neg W_{3,1}$... (step 4)

⊥ Contradiction found in 1 steps!

✓ SAFE DEDUCED: [1,2] — $\neg P \wedge \neg W$ proved via resolution